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The LQG Control Algorithms for Nonlinear Dynamic Systems

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Abstract

In this paper I will discuss some theoretical and practical aspects of the linear–quadratic–Gaussian (LQG) control algorithms implementation. The LQG control problem is one important optimal control problem and it can be used for systems where additive Gaussian noise appears on the system (output and state variables) or where the variable states are not complete measurable. The LQG control is a combination between one state estimator algorithm and one linear–quadratic regulator (LQR) algorithm. If the system to be controlled can be characterized by a linear system model, then the linear Kalman filter can be used. If the system is nonlinear, the Extended Kalman Filter (EKF) can be applied or other substitute methods such as the Unscented Kalman filter (UKF) algorithm.

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1. Introduction

For many various control algorithms one of the requirements is that the entire state vector of the controlled systems has to be accessible for measurement. In practice, the output of the process is most often measurable and very rarely can be accessed the state variables. In this case the state estimation process often plays an important role in the process control implementation. Industrial processes have many confusing factors that affect process control, such as model and measurement uncertainties, as well as many different unknown sources of noise. The well-known quadratic optimum control techniques can only be applied successfully if the model of the controlled process is

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known and all states are accessible. If the system model is linear it is one LQR (Linear Quadratic Regulator) control problem [1], [2]. If the system model is nonlinear, then this nonlinear model can be approximated with different simple, linear models, but this models are valid only around of the operating points. If there the controlled process is influenced with Gaussian additive noises, there can be used some filter algorithms to reduce the effect of these. The LQG control is a combination between one state estimator algorithm and one LQR algorithm. [4] If the studied system can be described with a nonlinear dynamic model than the most commonly used algorithm for the state estimation is the Extended Kalman Filter (EKF). Another solution would be to use the Unscented Kalman Filter (UKF) method. This is a derivative free method, and it resolves this problem by using a deterministic sampling approach.

This document is organized as follows. In Section 2, the theoretical approach to the DLQR control algorithm (state feedback control and tracking control) is presented. Part 3 contains some discrete state estimation algorithms. The validity of the presented approach is illustrated by numerical simulation results in section 4 using the nonlinear dynamical model of the inverted pendulum. Conclusions remarks are given in section 5.

2. Discrete time LQR algorithm

The optimal LQR control theory is applicable for one controllable dynamical system, what can be characterized by discrete time state space mathematical model. This mathematical representation can be written by the following general mathematical form:

$$\begin{aligned}x_k &= F(x_{k-1}, u_{k-1}) \\ y_k &= G(x_k, u_k)\end{aligned}\quad (1)$$

where $x_k \in \mathfrak{R}^{nx1}$ is the state vector, $u_k \in \mathfrak{R}^{mx1}$ is the input vector, $y_k \in \mathfrak{R}^{px1}$ is the output vector. The functions $F(): \mathfrak{R}^{n+m} \rightarrow \mathfrak{R}^n$ and $G(): \mathfrak{R}^{n+m} \rightarrow \mathfrak{R}^p$ are nonlinear and they need to be continuous. The LQR theory was developed for linear mathematical model. In the literature there are various approaches like identification, linearization or SDRE (State Dependent Riccati Equation) method can be used to determine the linearized mathematical model. The SDRE approach is applicable only for one special form of the nonlinear system model [3].

The nonlinear optimal controls in general include Jacobian linearization of the model system around of the each operating point. In this paper I assume that the nonlinear mathematical model of the studied system is known, and by linearization around its several operating points (x_k, u_k) different and simpler local mathematical models are obtained. The general form of the linearized mathematical model is formulated as:

$$\begin{aligned}x_{k+1} &= \Phi_k \cdot x_k + \Gamma_k \cdot u_k, \\ y_k &= C_k \cdot x_k\end{aligned}\quad (2)$$

$\Phi_k \in \mathfrak{R}^{nxn}$, $\Gamma_k \in \mathfrak{R}^{nxm}$ and $C_k \in \mathfrak{R}^{pxn}$ are matrices. The state space matrices is determined with the following approximations

$$\Phi_k \cong \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \dots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial x_1} & \dots & \frac{\partial F_n}{\partial x_n} \end{bmatrix}_{x_k, u_k} \quad \Gamma_k \cong \begin{bmatrix} \frac{\partial F_1}{\partial u_1} & \dots & \frac{\partial F_1}{\partial u_m} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial u_1} & \dots & \frac{\partial F_n}{\partial u_m} \end{bmatrix}_{x_k, u_k} \quad C_k \cong \begin{bmatrix} \frac{\partial G_1}{\partial x_1} & \dots & \frac{\partial G_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial G_p}{\partial x_1} & \dots & \frac{\partial G_p}{\partial x_n} \end{bmatrix}_{x_k, u_k} \quad (3)$$

This linear approximation of the model may introduce errors, so in this case the model based control algorithms is also can be present some incompleteness [1][2].

2.1. The state feedback control for discrete time LTI systems

The cost function to be minimized is given by the following relationship:

$$J(u) = \frac{1}{2} x_N^T \cdot F \cdot x_N + \frac{1}{2} \sum_{k=0}^{N-1} (x_k^T \cdot Q_k \cdot x_k + u_k^T \cdot R_k \cdot u_k) \quad (4)$$

where N is the horizon. The F , Q and the R are symmetric, positive weights matrices. The linear control law is computed from the following expression:

$$u_k = -\left(R_k + \Gamma_k^T \cdot P_{k+1} \cdot \Gamma_k\right)^{-1} \Gamma_k^T \cdot P_{k+1} \cdot \Phi_k \cdot x_k = -K_{fbk} \cdot x_k, \quad (5)$$

where P_{k+1} is the Riccati matrix which can be calculated solving the following discrete time Riccati equation:

$$P_k = \Phi_k^T \cdot P_{k+1} \cdot \Phi_k + Q_k - \Phi_k^T \cdot P_{k+1} \cdot \Gamma_k \left(R_k + \Gamma_k^T \cdot P_{k+1} \cdot \Gamma_k\right)^{-1} \Gamma_k^T \cdot P_{k+1} \cdot \Phi_k, \quad (6)$$

with the final condition $P_N = F$. If the system is linear and time invariant, assuming that $N \rightarrow \infty$ and Q and R are constant weighting matrices, then the Riccati matrix is the unique solution of the following discrete time algebraic Riccati equation:

$$P = \Phi^T \cdot P \cdot \Phi + Q - \Phi^T \cdot P \cdot \Gamma \cdot \left(R + \Gamma^T \cdot P \cdot \Gamma\right)^{-1} \Gamma^T \cdot P \cdot \Phi \quad (7)$$

Tuning of the LQR controller is performed by selecting these weighting matrices.

2.2. The tracking control for discrete time LTI systems

The cost function to be minimized is given by the following relationship:

$$J(u) = \frac{1}{2} e_N^T \cdot F \cdot e_N + \frac{1}{2} \sum_{k=0}^{N-1} (e_k^T \cdot Q_k \cdot e_k + u_k^T \cdot R_k \cdot u_k) \quad (8)$$

where e_k is the tracking error, difference between measured output and the reference signal. In this case the discrete time Riccati equation will be:

$$P_k = \Phi_k^T \cdot P_{k+1} \cdot \Phi_k + C_k^T \cdot Q_k \cdot C_k - \Phi_k^T \cdot P_{k+1} \cdot \Gamma_k \left(R_k + \Gamma_k^T \cdot P_{k+1} \cdot \Gamma_k\right)^{-1} \Gamma_k^T \cdot P_{k+1} \cdot \Phi_k \quad (9)$$

with the final condition $P_N = C^T \cdot F \cdot C$. If the system is time invariant, assuming that $N \rightarrow \infty$ and Q and R are constant weight matrices, then the Riccati matrix is the unique solution of the discrete time algebraic Riccati equation.

The linear control law can be determinate is computed from the following expression:

$$\begin{aligned} u_k &= K_{ff} \cdot z_k - K_{fb} \cdot x_k \\ K_{fb} &= \left(R + \Gamma_k^T \cdot P_k \cdot \Gamma_k\right)^{-1} \Gamma_k^T \cdot P_k \cdot \Phi_k \\ K_{ff} &= \left(R + \Gamma_k^T \cdot P_k \cdot \Gamma_k\right)^{-1} \Gamma_k^T \cdot \left(\Phi_k^T + K_{fb}^T \cdot \Gamma_k^T - I\right)^{-1} \cdot C_k^T \cdot Q \end{aligned} \quad (10)$$

where the K_{fb} is the feed-back and K_{ff} is the feed-forward gain vector.

3. The discrete time state estimation algorithm

We consider the following nonlinear discrete process model:

$$\begin{aligned}x_k &= F(x_{k-1}, u_{k-1}) + w_{k-1} \\ y_k &= G(x_k, u_k) + v_k\end{aligned}\quad (11)$$

where w_k is the process noise and v_k is the observation noise. Both noises are Gaussian, independent random processes with zero means and known time invariant covariance matrices:

$$\begin{aligned}Q_- &= \text{cov}\{w\} = E\{w \cdot w^T\} \\ R_- &= \text{cov}\{v\} = E\{v \cdot v^T\}\end{aligned}\quad (12)$$

where the $E\{\}$ is the expected value operator.

3.1. The EKF algorithm for nonlinear state estimation

The Kalman filter propagates the mean and covariance of the probability distribution function of the model state in an optimal way with minimization of the mean square error. The Kalman filter dynamics results from the consecutive cycles of prediction and filtering. The EKF algorithm for state estimation can be applied to the discrete time system model.

The steps of the EKF state estimation algorithm is presented below [5], [6] and [10]:

0. Initialization step at $k=0$.

- Initial estimated state vector $\tilde{x}_0 = E\{x_0\}$
- Covariance matrix: $\tilde{P}_{0,x} = E\{(x_0 - \tilde{x}_0) \cdot (x_0 - \tilde{x}_0)^T\}$

1. Linearization step ($k \geq 1$): linearizing the model (with the Taylor series expansion) we can calculate the following matrices:

$$\Phi_k = \begin{bmatrix} \frac{\partial F_1}{\partial x_1} & \dots & \frac{\partial F_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_n}{\partial x_1} & \dots & \frac{\partial F_n}{\partial x_n} \end{bmatrix}_{\tilde{x}_{k-1}, u_{k-1}} \quad C_k = \begin{bmatrix} \frac{\partial G_1}{\partial x_1} & \dots & \frac{\partial G_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial G_p}{\partial x_1} & \dots & \frac{\partial G_p}{\partial x_n} \end{bmatrix}_{\tilde{x}_{k-1}, u_{k-1}} \quad (13)$$

2. Prediction step: Calculation of the predicted state mean and covariance (time update)

$$\begin{aligned}\tilde{x}_k^- &= F(\tilde{x}_{k-1}, u_{k-1}) \\ \tilde{P}_{x,k}^- &= \Phi_k \cdot \tilde{P}_{x,k-1} \cdot \Phi_k^T + Q_-\end{aligned}\quad (14)$$

3. Calculation of the estimation filter gain vector:

$$K_k = \tilde{P}_{x,k}^- \cdot C_k^T \cdot (C_k \cdot \tilde{P}_{x,k}^- \cdot C_k^T + R_-)^{-1} \quad (15)$$

4. Correction step: The estimates are updated with latest observation (measurement update)

$$\begin{aligned}\tilde{x}_k &= \tilde{x}_k^- + K_k \cdot (y_k - G(\tilde{x}_k^-, u_k)) \\ \tilde{P}_{x,k} &= (I - K_k \cdot C_k) \cdot \tilde{P}_{x,k}^-\end{aligned}\quad (16)$$

The EKF algorithm gives an approximation of the optimal estimate. The non-linearities of the system's dynamics are approximated by a linearized version of the non-linear system model around the last estimated state. This algorithm can be divergent if the consecutive linearization are not a good approximation of the non-linear model.

3.2. The Unscented Transformation and UKF algorithm for nonlinear state estimation

Consider the non-linear system described by the equation (11). The state distribution is also represented by Gaussian random variables, but this method is using a minimal set of carefully chosen sample points. These points, called sigma points, completely capture the true mean and covariance of the states and are propagated through the nonlinearity.

For calculating the statistics of a random variable which undergoes a nonlinear transformation we can use the unscented transformation (UT) [7], [8]. Consider a random variable x (dimension n), which is propagating through a nonlinear function $y = f(x)$. The mean of x is $\bar{x} = E\{x\}$ and the covariance of x is $P_x = E\{(x - \bar{x}) \cdot (x - \bar{x})^T\}$. To calculate the statistics of y , we form a matrix $\mathbb{X}$ of $2 \cdot n + 1$ sigma vectors according to the following:

$$\begin{aligned}\mathbb{X}_0 &= \bar{x} \\ \mathbb{X}_i &= \bar{x} + \gamma \cdot \left(\sqrt{P_x}\right)_i \quad i = 1, \dots, n \\ \mathbb{X}_i &= \bar{x} - \gamma \cdot \left(\sqrt{P_x}\right)_i \quad i = n + 1, \dots, 2 \cdot n\end{aligned}\quad (17)$$

where $\gamma = \sqrt{n + \lambda}$ is a scaling parameter and $\lambda = \alpha^2 \cdot (n + k) - n$. The constant α determines the spread of the sigma points around the mean value $\bar{x}$ and usually $\alpha \in [10^{-4}, 1]$. The constant k is a secondary scaling parameter and it is usually set to 0 for state estimation. $\left(\sqrt{P_x}\right)_i$ is the i -th column of the matrix square root. These $\mathbb{X}_i$ vectors are propagated through the nonlinear function

$$y_i = f(\mathbb{X}_i) \quad i = 0, \dots, 2 \cdot n \quad (18)$$

The mean and covariance for y are approximated using a weighted sample mean and covariance of the posterior sigma points:

$$\begin{aligned}\bar{y} &\approx \sum_{i=0}^{2 \cdot n} W_i^{(m)} \cdot y_i \\ P_y &\approx \sum_{i=0}^{2 \cdot n} W_i^{(c)} \cdot (y_i - \bar{y}) \cdot (y_i - \bar{y})^T\end{aligned}\quad (19)$$

The weights are given by

$$\begin{aligned}W_0^{(m)} &= \lambda / (n + \lambda) \\ W_0^{(c)} &= \lambda / (n + \lambda) + (1 - \alpha^2 + \beta) \\ W_i^{(m)} &= W_i^{(c)} = 1 / (2 \cdot (n + \lambda)) \quad i = 1, \dots, 2 \cdot n\end{aligned}\quad (20)$$

where β is used to incorporate prior knowledge of the distribution of x (for Gaussian distribution the value is 2).

The standard UKF state estimation algorithm, with additive (zero mean) noise, is presented below [9], [10], [5]:

0. Initialization step at $k=0$:

- initial value of the estimated state vector $\tilde{x}_0 = E\{x_0\}$
- initial value of the covariance matrix: $\tilde{P}_{x,0} = E\{(x_0 - \tilde{x}_0) \cdot (x_0 - \tilde{x}_0)^T\}$.

1. Sigma points' calculation for $k \geq 1$

$$\mathfrak{S}_{k-1} = \begin{bmatrix} \tilde{x}_{k-1} & \tilde{x}_{k-1} + \gamma \cdot \sqrt{\tilde{P}_{x,k-1}} & \tilde{x}_{k-1} - \gamma \cdot \sqrt{\tilde{P}_{x,k-1}} \end{bmatrix} \quad (21)$$

2. Propagation of the sigma points:

- transform the sigma points through the state-update function

$$\mathfrak{S}_k^* = F(\mathfrak{S}_{k-1}, u_{k-1}) \quad (22)$$

- calculate the apriori state estimate and apriori covariance matrix:

$$\begin{aligned} \tilde{x}_k^- &= \sum_{i=0}^{2n} W_i^{(m)} \cdot (\mathfrak{S}_k^*)_i \\ \tilde{P}_{x,k}^- &= \sum_{i=0}^{2n} W_i^{(c)} \cdot ((\mathfrak{S}_k^*)_i - \tilde{x}_k^-) \cdot ((\mathfrak{S}_k^*)_i - \tilde{x}_k^-)^T + Q_- \end{aligned} \quad (23)$$

The weights $W_i^{(m)}$ and $W_i^{(c)}$ are defined in accordance with relations (20).

3. Update of the output vectors:

- transform the sigma points through the measurement-update function

$$Y_k^* = G(\mathfrak{S}_{k-1}, u_k) \quad (24)$$

- calculate the mean and covariance of the measurement vector

$$\begin{aligned} \tilde{y}_k^- &= \sum_{i=0}^{2n} W_i^{(m)} \cdot (Y_k^*)_i \\ \tilde{P}_{y,k}^- &= \sum_{i=0}^{2n} W_i^{(c)} \cdot ((Y_k^*)_i - \tilde{y}_k^-) \cdot ((Y_k^*)_i - \tilde{y}_k^-)^T + R_- \end{aligned} \quad (25)$$

4. Calculate the cross covariance matrix:

$$\tilde{P}_{x,y,k} = \sum_{i=0}^{2n} W_i^{(c)} \cdot ((\mathfrak{S}_k^*)_i - \tilde{x}_k^-) \cdot ((Y_k^*)_i - \tilde{y}_k^-)^T \quad (26)$$

5. Calculation of the Kalman filter gain vector

$$K_k = \tilde{P}_{x,y,k} \cdot (\tilde{P}_{y,k}^-)^{-1} \quad (27)$$

6. Calculate the estimated state and the covariance in accordance with the standard Kalman filter algorithm:

$$\begin{aligned}\tilde{\underline{x}}_k &= \tilde{\underline{x}}_k^- + K_k \cdot (y_k - \tilde{y}_k^-) \\ \tilde{P}_{x,k} &= \tilde{P}_{x,k}^- - K_k \cdot \tilde{P}_{y,k}^- \cdot K_k^T\end{aligned}\quad (28)$$

The UKF principle is simple and easy to implement because it does not require the calculation of Jacobians at each time step. The most computationally intensive operation in the UKF corresponds to calculating the new set of sigma points at each time update.

4. Numerical simulation. Example.

For testing the presented control and state estimation algorithms, I used the well-known inverse pendulum dynamical system. The controlling of the inverted pendulum on a cart is quite difficult. This problem is due to the different characteristics of the system: instability, nonlinearity of the model, with single input and two output signals (underactuated) [11]. The schematic representation of the inverted pendulum system is shown in Fig. 1.

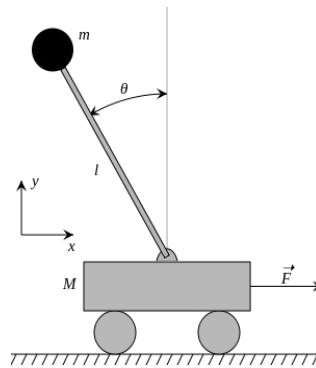


Fig. 1. Inverted pendulum on cart

The investigated system is characterized by the following nonlinear continuous time state space model [11] [12].

$$\begin{aligned}\dot{x}_1(t) &= x_2(t) \\ \dot{x}_2(t) &= \frac{(l \cdot x_4^2(t) \cdot \sin(x_3(t)) - g \cdot \sin(x_3(t)) \cdot \cos(x_3(t)) + \frac{1}{m} \cdot u(t))}{(M/m + \sin^2(x_3(t)))} \\ \dot{x}_3(t) &= x_4(t) \\ \dot{x}_4(t) &= \frac{\left(\frac{(m+M)}{m \cdot l} g \cdot \sin(x_3(t)) - x_4^2(t) \cdot \sin(x_3(t)) \cdot \cos(x_3(t)) - \frac{1}{m \cdot l} \cdot \cos(x_3(t)) \cdot u(t) \right)}{(M/m + \sin^2(x_3(t)))}\end{aligned}\quad (29)$$

The four state variables in the model are:

- $x_1(t) = x(t)$ - displacement of cart;
- $x_2(t) = \dot{x}(t)$ - speed of cart;
- $x_3(t) = \theta(t)$ - pendulum angle;
- $x_4(t) = \dot{\theta}(t)$ - pendulum angle speed.

In this model the inertia of the system is neglected and the Euler-Lagrange equations theory was used to determine this mathematical model. I implemented the state control DLQR algorithm where the goal is to set the entire states to zero. This means that the output signal is not important just for the control process. In this case the reference signal set to zero. The output signals will be considered $y_1(t) = x_1(t)$ displacement of the cart and $y_2(t) = x_3(t)$ (pendulum angle). It is very important the selection correctly of the output signals. In order for the state

estimator to work properly, the system must be observable. Process and measurement noises with known variances are added to the system states and measurements.

The Fig. 2 illustrates the schematic representation of the implemented simulation applications. The controlled plant is the dynamical state space model of the inverted pendulum. The parameters in the model and their numeric values are shown in Table 1. In this table I also show the simulation parameters used for testing the presented algorithms. I have also outlined in this table the values of the most important control and estimation parameters (this parameters have strongly effects to the convergence of the control and estimation algorithms).

In all the simulations have been made in the similar conditions and I used the parameters illustrated in Table 1.

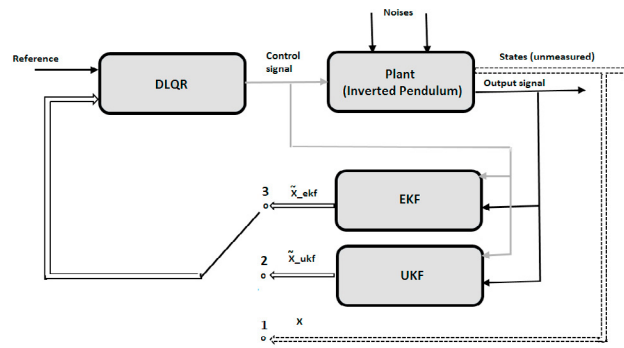


Fig. 2. Schematic representation of the implemented algorithms.

First, I only tested the DLQR control algorithm, in this case I assumed that was no noise on the system and all the state of the system was available (measurable). The graphical results can be seen at Fig. 3, the control results (controlled states and the control signal) in this case the state estimator algorithms do not work. The goal is to control the states to zero, so when testing the algorithm, I change the state values two times in simulation. At first I change the position of the cart ($t \sim 14$ sec) and later change the angle of the pendulum ($t \sim 27$ sec).

Table 1. Parameters.

System parameters	Value
M (mass of cart,)	0.6 kg
m (mass of pendulum)	0.3 kg
l (length of pendulum)	0.35 m
g (acceleration of gravity)	9.81 m/s ²
Simulation and DLQR control parameters	
Ts (sample time)	0.02 sec
x0 (initial state)	[-1 0 0.01 0]
R (weighting matrix of the controls)	0.1
Q (weighting matrix of the states)	100 I ₄
Estimation parameters (EKF, UKF)	
x0 [~] (initial estimated state vector)	[-0.9 0.5 0.21 -0.1]
P0 (initial covariance matrix)	100 I ₄
Q ₋ (covariance matrix, process noise)	0.01 I ₄
R ₋ (covariance matrix, measurement noise)	0.1 I ₂

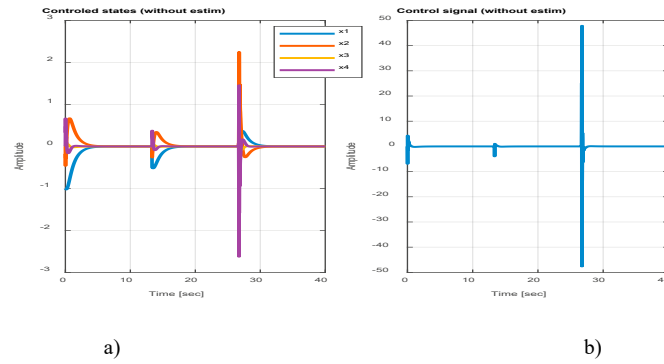


Fig. 3. The controlled states (a) and the control signal (b) without any state estimation algorithms.

The estimation algorithms were tested first also without noises, and were compared the estimated states with the ideal (non-measurable) values (Fig. 5). For both estimator algorithms, I got similar results, so here I present just the EKF algorithms results (Fig. 4 and Fig. 5).

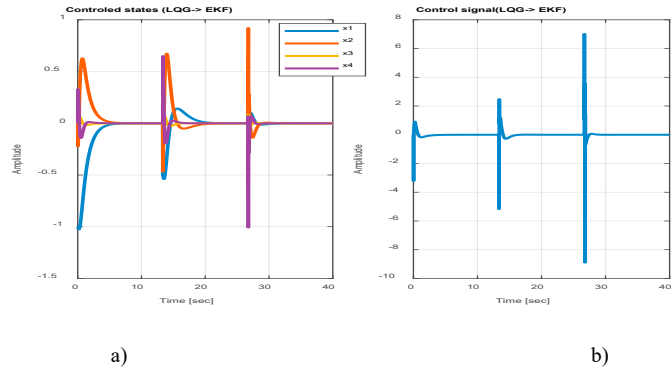


Fig. 4. The controlled states (a) and the control signal (b) with EKF estimation algorithms (without noise)

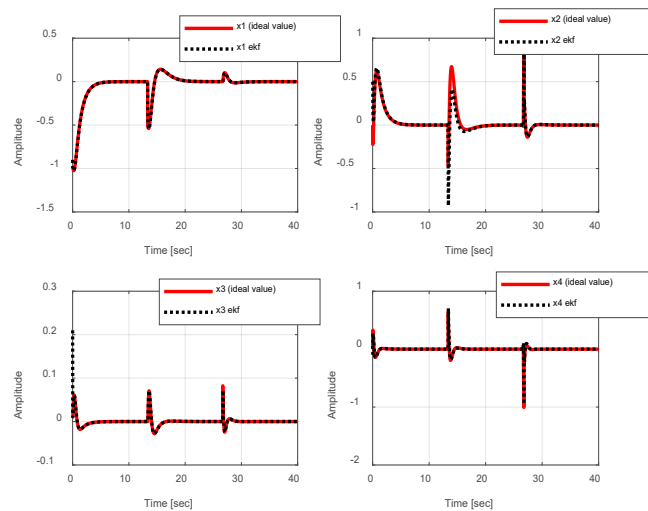


Fig. 5. Comparison between estimated states and ideal (non-measurable) values.

Finally I tested LQG algorithms in the presence of external Gaussian noise. The graphical results (now using the UKF method) are given in Figures 6 and 7. The LQG regulatory results are also shown here, now the UKF algorithm is in operation (Fig. 6). Here the algorithm's filter effect can also be seen, showing the noisy measured outputs with the estimated (filtered) output signals (Fig. 7).

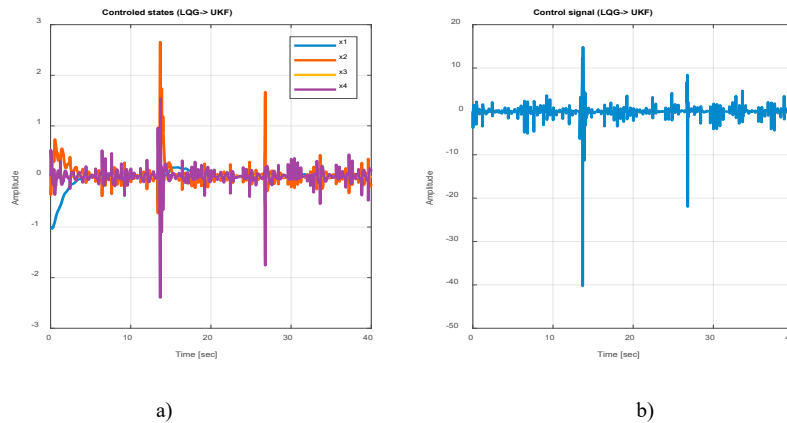


Fig. 6. The controlled states (a) and the control signal (b) with UKF estimation algorithms (with noise).

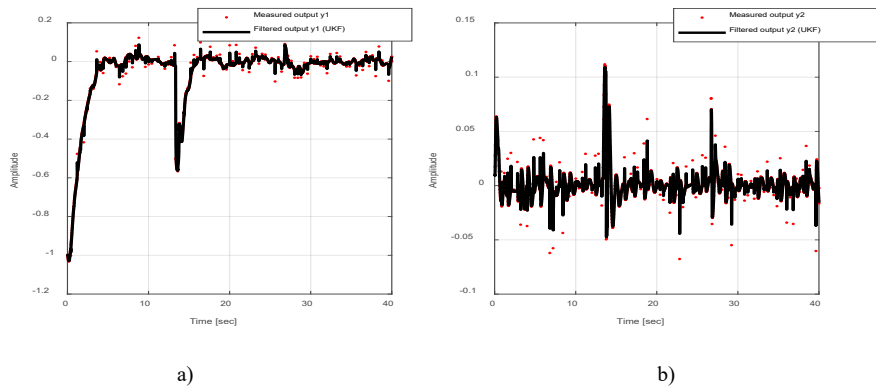


Fig. 7. The measured and estimated output signals with UKF algorithm. (a) displacement of cart; (b) pendulum angle.

I mentioned that the graphical results of the two estimation algorithms are very similar. Therefore I made a comparison for the change in the trace of covariance matrix, and calculated the sum of the square errors in both 2 method. The graphical results are shown in Fig. 8. It can be seen that both methods are well suited for state estimation, but with the UKF method, we get a smaller estimation error result.

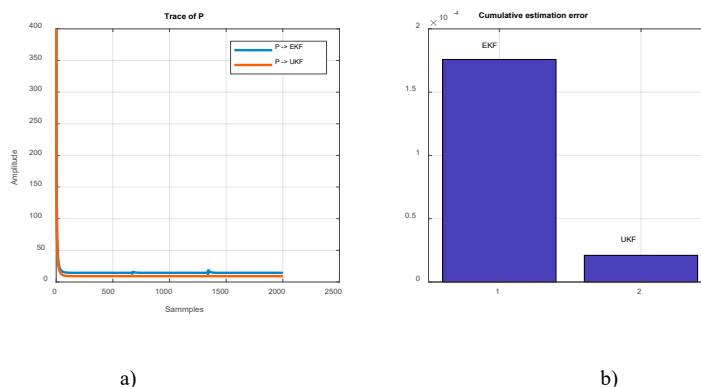


Fig. 8. Trace of covariance matrix (a) and cumulative estimation error (b)

4. Conclusions

This paper introduces the theory of LQG control and its application on nonlinear systems. The LQG control is a combination between one state estimator algorithm and one linear–quadratic regulator (LQR) algorithm. Two versions of the state estimation were also presented: the EKF and the UKF algorithms. The UKF algorithm for state estimation is an interesting alternative to the EKF because it has an improved performance. The theoretical algorithms were tested on a nonlinear system (inverted pendulum) and then they were compared..

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